

PDL TUTORIAL

Dr. Ume Hani

**TUTORIAL TASK 1: IDENTIFY,
STUDY AND LABEL THE GIVEN
DIAGRAMS.**

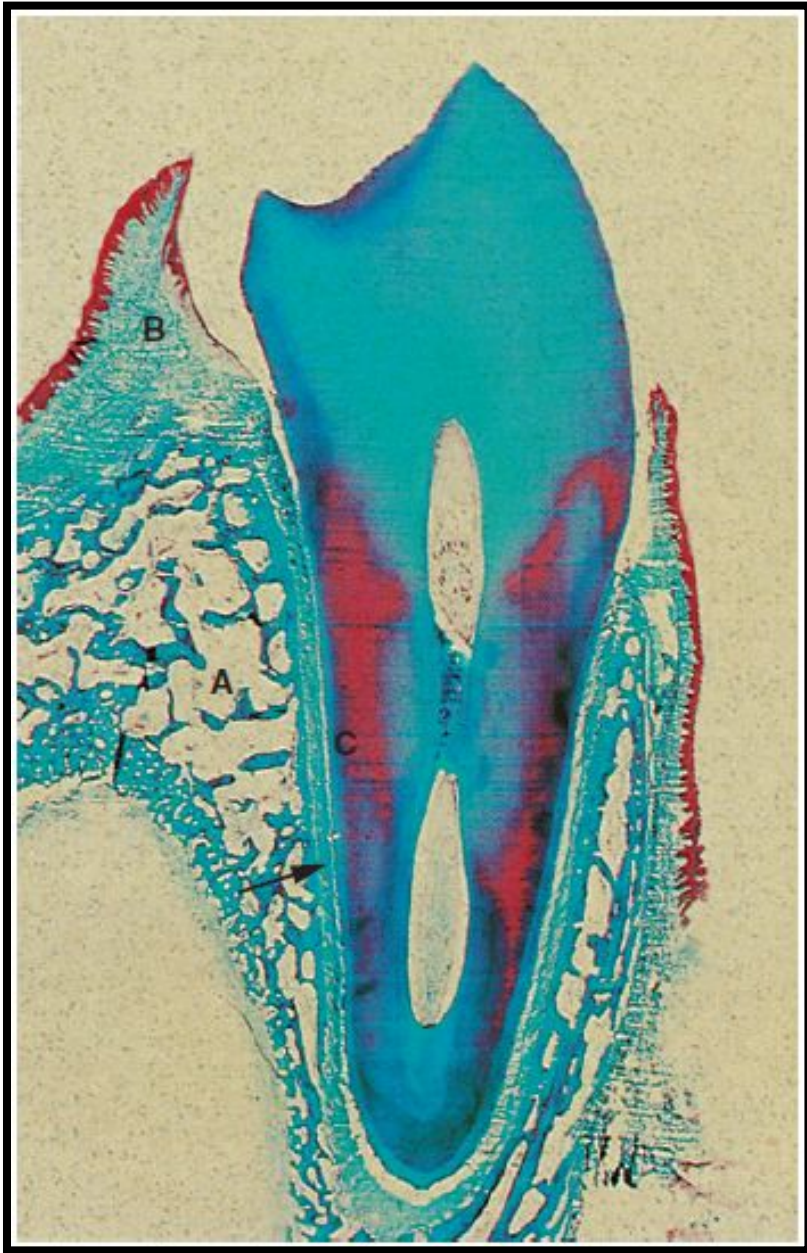


Fig. 12.1 The relationship of the periodontal ligament and the periodontal space (arrow) to the other tissues of the periodontium.

A = alveolar bone

B = gingiva

C = root lined by cementum

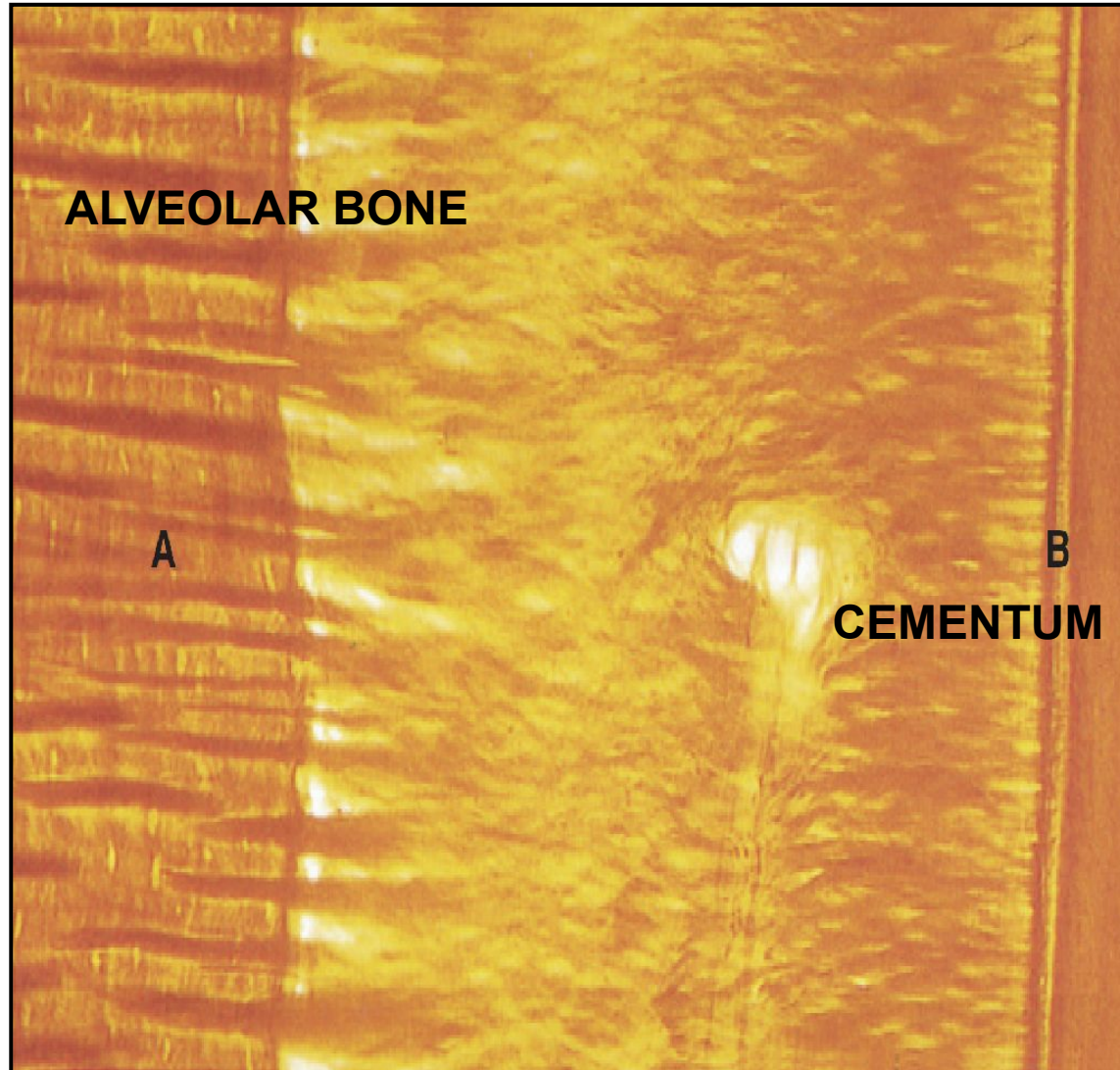


Fig. 12.19 The insertion of periodontal fibres into alveolar bone (A) and cementum (B). The horizontal lines in the bone and cementum represent the Sharpey's fibres.

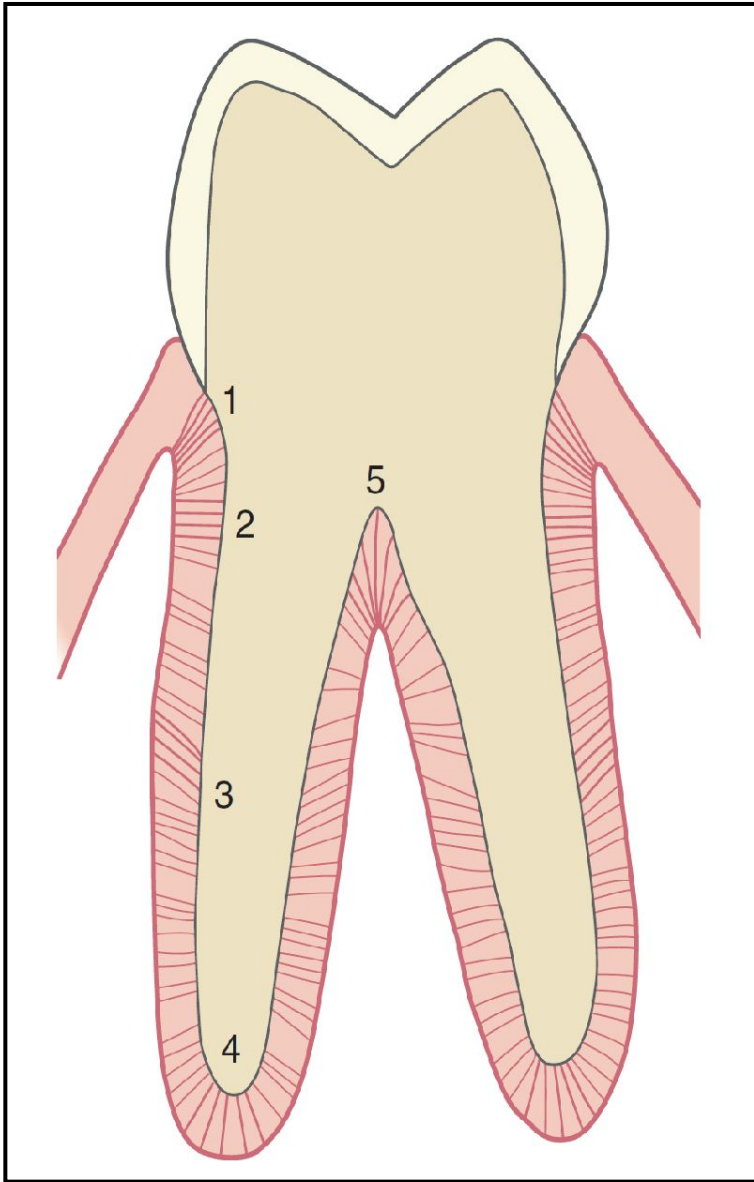


Fig. 12.9: The orientation of the principal fibres of the periodontal ligament seen in longitudinal section of a multirooted tooth:

1 = dentoalveolar crest fibres;

2 =horizontal fibres;

3 = oblique fibres;

4 = apical fibres;

5 = interradicular fibres.

TUTORIAL TASK 2: QUESTIONS AND ANSWERS

Q1- ENUMERATE THE PRINCIPAL FIBERS OF PDL AND THEIR LOCATION?

Principal Fibers of Fiber	Location
Dentoalveolar Crest Fibers	- Run from the cementum just beneath the alveolar crest to the alveolar bone.
Horizontal Fibers	- Extend horizontally from the cementum to the alveolar bone, just apical to the crest fibers.
Oblique Fibers	- Run obliquely (in a downward direction) from the cementum to the alveolar bone, supporting the tooth against masticatory forces.
Apical Fibers	- Extend from the apex of the tooth's cementum down to the alveolar bone, surrounding the tooth apex.
Interradicular Fibers	- Located only in multi-rooted teeth, these fibers extend from cementum between the roots of the teeth to the interradicular septum of the alveolar bone.

Q2- Define Sharpey's fibers and what is difference between The Sharpey's fibers OF alveolar bone and that of cementum?

- The principal fibres of the periodontal ligament that are 'embedded' into cementum and the bone lining the tooth socket are termed Sharpey's fibres.**
- These Sharpey's fibres are different when attached into either side of alveolar bone and cementum since, they are more numerous, but smaller, at their attachments into cementum than at the alveolar bone.**
- They are fully mineralized at the level of insertion in alveolar bone and partially mineralized in cementum.**